

Annabel Li – Engineering Physics

li.annabel47@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | (778)-990-0188

PROFILE

UBC Engineering Physics student eager to tackle R&D challenges at the intersection of electrical, software, and mechanical engineering. Skilled in machine learning, data analysis and mechanical design, with a record of success in individual and team-based projects.

SKILLS

Software: Machine Learning (Tensorflow, Keras); Python, C/C++; Git

Engineering design: CAD (SolidWorks); CFD (Star CCM+); Vitis, Vivado; machining techniques

Electronics: Oscilloscopes, digital multimeters, circuits

Relevant Coursework: UBC PHAS Student Machine Shop Course (waterjet, lathe, milling, hand tools)

TECHNICAL EXPERIENCE

Deep Learning Student Research Assistant, Atlas Canada May – Sept. 2025

TRIUMF Particle Accelerator – First Year Summer Research Experience Award Vancouver, BC

- **Optimized machine learning models** for high data-pileup environments using Tensorflow and Keras, reducing inference time by **20-30%** and model size by **50-60%**
- **Developed automated Python workflows** to train, evaluate, and debug models, reducing experiment turnaround time and enabling **4x more results** per day
- **Built modular neural networks in C++** for FPGA deployment with Vitis and Vivado, designing data processing pipelines to format numbers in hardware-compatible forms
- **Presented research findings** at TRIUMF'S Science Week, earning **third place** among all undergraduate and graduate student presenters

Aerodynamics member, UBC Formula SAE

Sept. 2024 – present

University of British Columbia

Vancouver, BC

- **Led the end-to-end design of the beam wing** for the 2025 vehicle, overseeing research, simulation, and documentation
- **Improved aerodynamic balance by 2-3%**, resulting in consistent driver reports of improved vehicle handling
- Conducted iterative testing with **CAD** (Solidworks) and **CFD** (Star CCM+) to integrate feedback and refine designs based on team findings
- Manufactured the final component with machining and composite techniques

HONOURS AND AWARDS

UBC Trek Excellence Scholarship (2025) – *awarded to top 5% domestic students in faculty*

First Year Summer Research Experience Award, University of British Columbia (2025)

C. K. Choi Entrance Scholarship, University of British Columbia (2024)

EDUCATION

Bachelor of Applied Science, Engineering Physics (in-progress)

University of British Columbia

Most recent average: 93%

EXTRACURRICULAR EXPERIENCE

- Royal Conservatory of Music Pianist (2011 – 2024)
- FIS (internationally-certified) ski racer (2022 – 2024)